Tracking neurons across days with high-density probes

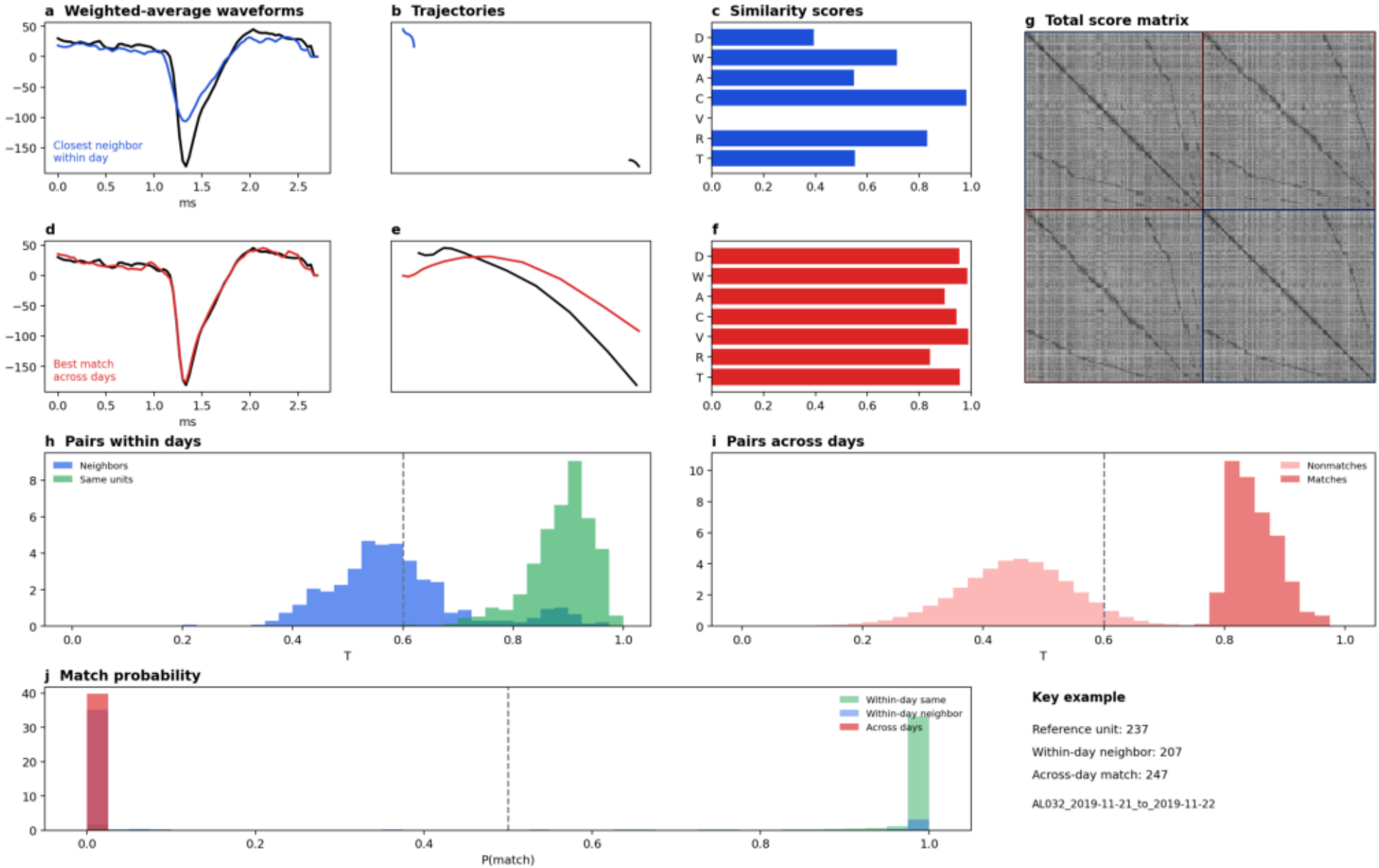
This is a local full-data UnitMatch report assembled in a paper-like format from the bundled recordings.

It consolidates the existing exploratory outputs into a single clean package and focuses on the figure families requested for review: classifier construction, functional validation, and tracking over many recordings.

Natural-image-response validation from the published paper is not reproducible from the files present in this checkout, so the functional-validation section replaces that portion with a spike-time-derived reference-population analog.

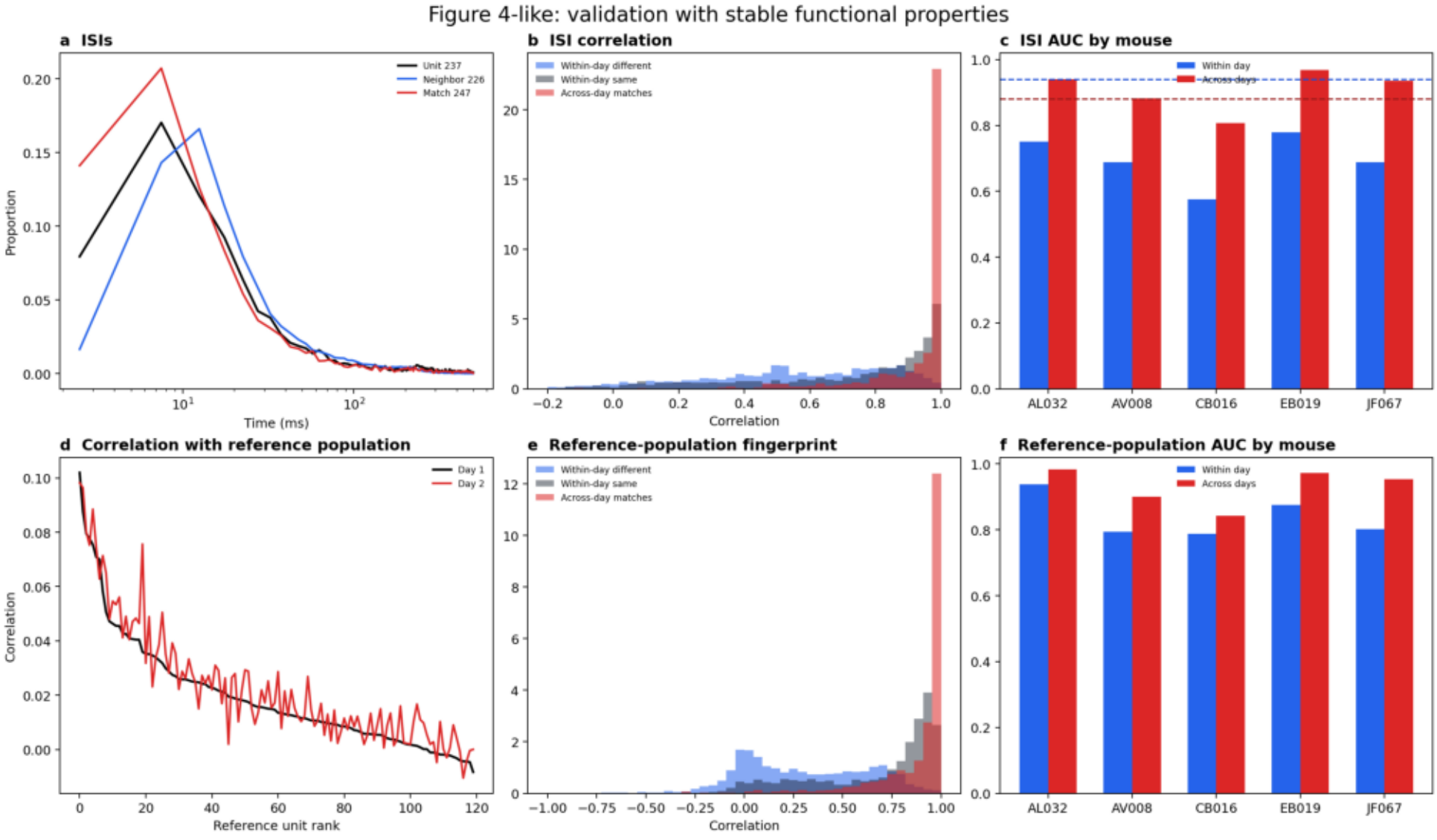
Metric	Value
Sessions processed	20
Pairs processed	10
Mice	AL032, AV008, CB016, EB019, JF067
Acute mean tracked fraction	40.7%
Chronic mean tracked fraction	46.3%
Within-day false-positive median	0.19%
Within-day false-negative median	9.25%
ISI across-day AUC mean	0.906
Reference-population across-day AUC mean	0.930

Figure 3-like: computing similarity scores and setting up the classifier



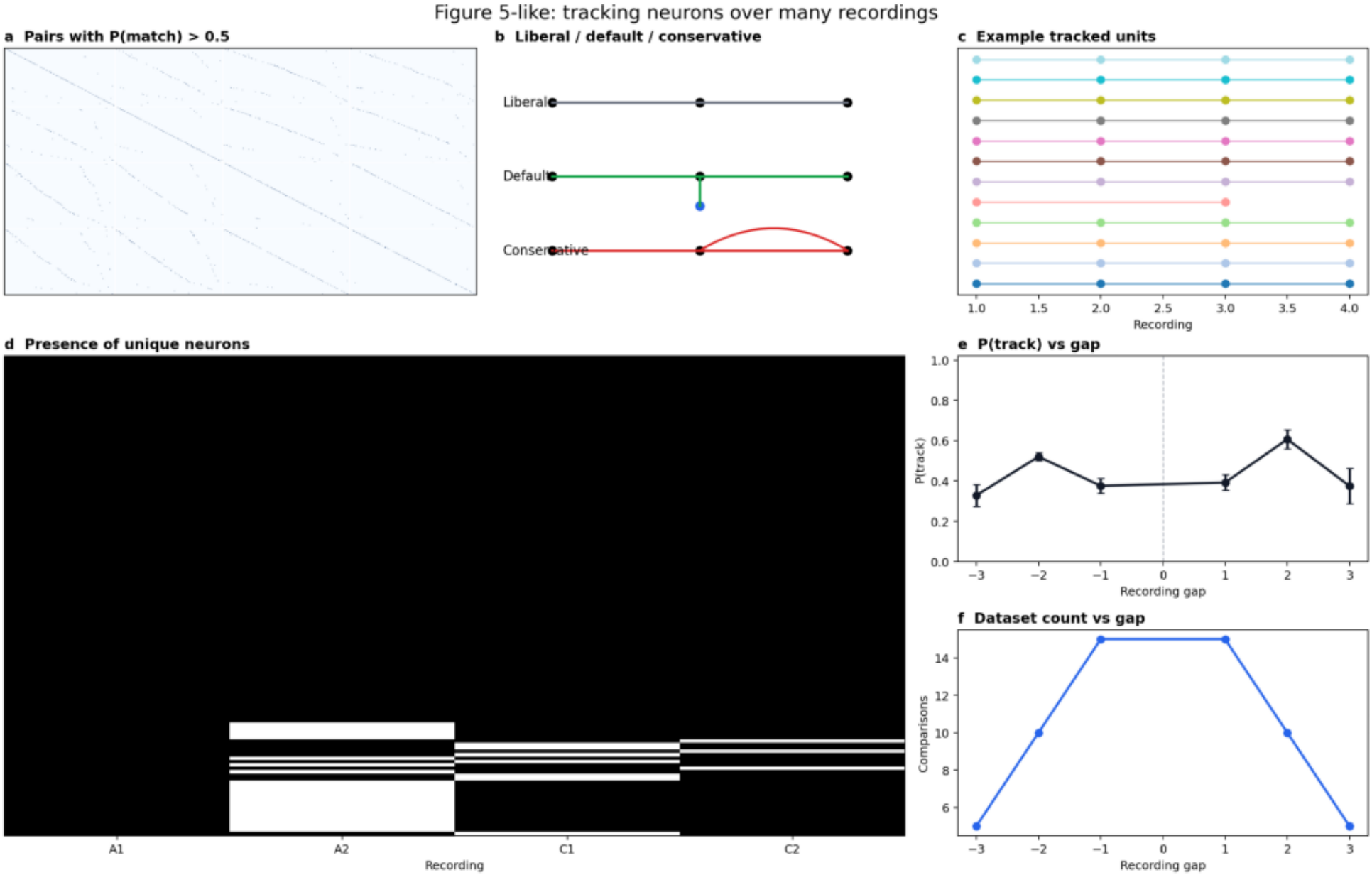
Local analog built from the AL032 consecutive-day pair.

Figure 4-like: validation with stable functional properties



Local analog using ISI histograms and a spike-time-derived reference-population fingerprint.

Figure 5-like: tracking neurons over many recordings



Local analog using the saved multi-session classic runs across the bundled mice.